Car Fuel Economy Analysis

1.Project Overview and Objective:

- This project involves cleaning, transforming, and analysing raw data using python and creating an interactive dashboard to derive meaningful business insights.
- The main objective is to demonstrate data pre-processing techniques using python and an interactive dashboard visualization to make informed decisions.

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2. Data Sources

- **Source Description :** Public automobile datasets compiled from open data portals such as the UCI Machine Learning Repository and vehicle fuel economy data similar to records published by the United States Environmental Protection Agency.
- **Domain**: Automobile

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3. Problem Statement

- Identify which **car brands** and **models** have **higher fuel efficiency**
- Determine which cars have **better sales performance**
- Evaluate **the overall performance of cars** by combining both **fuel efficiency** and **sales efficiency**

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4. Attribute (Column /Features) Details:

Attribute Name	Data Type	Description
Car_ID	String	Car ID is a unique identifier assigned to each car in the dataset.
Brand	String	Name represents the company name of the car in the dataset
Model	String	Represents the specific model name of the car produced by a manufacturer. Each model identifies a particular design or version of a vehicle
year	String	Indicates the manufacturing or release year of the car model. This helps analyze trends in vehicle performance across different years.

City_MPG	Numerical	City Miles Per Gallon (MPG) represents the fuel efficiency of a car when driven in city conditions with frequent stops and traffic. Higher values indicate better fuel economy in city driving.
Highway_MPG	Numerical	Highway Miles Per Gallon (MPG) indicates the fuel efficiency of the car when driven on highways at steady speeds. Higher MPG means the vehicle consumes less fuel during long-distance travel.
Sales_Efficiency	Numerical	Represents the sales performance of the car model in the market. Higher values indicate higher demand or better sales performance..
Fuel_Efficiency	Numeric	Indicates the overall fuel economy performance of the vehicle. It is usually derived from city and highway fuel efficiency values to measure how efficiently the car uses fuel.
Cylinders	Numeric	Represents the number of cylinders in the car's engine. More cylinders generally provide higher power but may consume more fuel.
Engine_Size	Categorical	Indicates the size of the car engine, usually measured in liters (L). Larger engine sizes typically produce more power but may reduce fuel efficiency.
Transmission	Numeric	Describes the type of transmission system used in the vehicle, such as **Automatic** or **Manual**, which

		affects driving experience and fuel consumption.
Fuel_Type	Categorical	Indicates the type of fuel used by the vehicle, such as **Petrol, Diesel, Hybrid, or Electric**. This helps analyze fuel efficiency across different fuel types.
Sales_Category	Numeric	Classifies cars based on their sales performance, such as **Low Sales, Medium Sales, or High Sales**, to easily compare market demand across models

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5. Tools & Technologies

Python & PowerBI: Data cleaning, transformation, Pivot Tables, Data modelling, calculations, visualization, and interactive dashboard creation.

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6. Data Pre-Processing (Python)

Tasks Performed:

- **Data Cleaning & Transformation:**
- **Using Pandas, Numpy, dataframe**
- Created calculated field to find the fuel efficiency status of each car.
- **Filtering & Sorting:** Data was filtered to find fuel demands overall in each brand.
- **Pivot Tables:** Generated Pivot Tables for data summarization and initial insights.
    1. Created Pivot Tables for Maximum sales efficiency by brand,model,year.
    2. Created Pivot table to analyse fueltype demands by user.
- **Min/Max Scaler:** Min/Max scaler to find create overall sales and fuel efficiency by car, model, year.

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8. EDA and Visualizations

Dashboard Features:

Using matplotlib & seaborn

- **Multiple Visualizations based on problem statement:** Bar charts, line charts, pie charts, and tables to communicate insights.

  1. Bar chart to find car wise fuel variation.

  2. Pie chart for car vs Sales.

  3. Histogram for fuel efficiency distribution.

4. Count plot for compare high, medium, low & poor sales

5. Heatmap to find correlation.

6. Scatter plot to find impact of engine size with fuel efficiency.

- **Created a consolidated Report /Dashboard.**
- **Descriptive Analysis**
- **Diagnostic Analysis**
- **Predictive Analysis**
- **Prescriptive Analysis**

Consolidated Refinery Insights Report

9. Insights & Conclusions

- **Key Findings:**
- **Fuel Efficiency Variation Across Models**:
  The analysis shows that fuel efficiency varies significantly across different car models and brands. Some models provide higher City MPG and Highway MPG, indicating better fuel economy.

Relationship Between Engine Size and Fuel Efficiency:
Cars with larger engine sizes and more cylinders generally tend to have lower fuel efficiency, while smaller engines often provide better mileage.

Impact of Transmission Type:
Vehicles with automatic and manual transmissions show differences in fuel consumption. In some cases, manual transmission cars demonstrate slightly better fuel efficiency.

Sales Performance of Car Models:
Certain car models show higher sales efficiency, indicating strong market demand. These models usually balance both performance and fuel economy.

Fuel Type Influence:
Vehicles using different fuel types such as petrol, diesel, or hybrid show variations in fuel economy. Hybrid or fuel-efficient vehicles tend to provide better mileage.

Overall Performance Evaluation:
By combining Sales Efficiency and Fuel Efficiency, the analysis helps identify the best-performing car models that balance market demand and fuel economy.

- ### 10. Conclusions
- The analysis of the car fuel economy dataset provides valuable insights into the relationship between **fuel efficiency, engine characteristics, and sales performance** of different car models. The results show that vehicles with **smaller engine sizes and fewer cylinders** generally achieve better fuel efficiency compared to larger engines.
- Additionally, the study highlights that some car models achieve a balance between **high sales efficiency and good fuel economy**, making them the best-performing vehicles in the market. By applying **data scaling and overall performance scoring**, it becomes easier to identify top-performing cars based on both market demand and fuel efficiency.

- Overall, this analysis helps understand how **vehicle specifications and fuel economy influence customer preferences and sales performance**